Quantitative Metrics for Decomposing Timing and Magnitude Differences in Influenza Forecasts

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Outline

Why Does Sub-State Surveillance Matter?

- Public health decisions are often made at the **state level**, but influenza dynamics vary dramatically within states.
- Understanding how local epidemic curves *deviate* from the state curve could improve:
  - Surge capacity planning
  - Resource allocation timing
  - Sub-state early warning

Research Question

Which curve similarity metric best captures the *timing* and *magnitude* differences between HSA-level and state-level influenza incidence?

Key Comparison

DTW and **Fréchet distance** vs. conventional RMSE

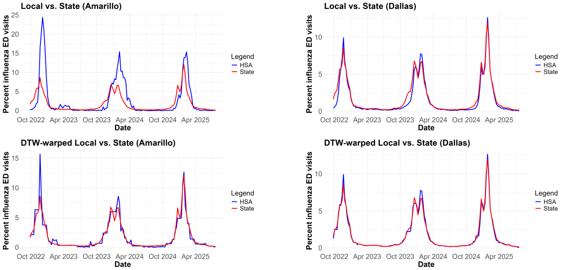
Data & Study Design

Seasons Analyzed

- 2022/23 · 2023/24 · 2024/25
- ∼148 total epidemic weeks

Geographic Unit

- ∼148 HSAs across the lower 48 states
- Each HSA paired with its parent state curve



Case Studies

Amarillo, TX: high divergence from Texas' state curve

Dallas, TX: low divergence from Texas' state curve

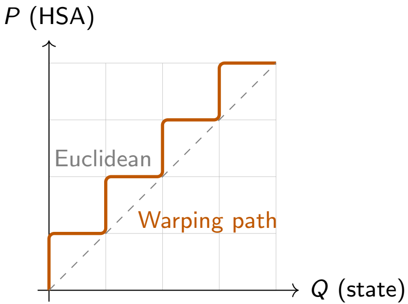
What Is Dynamic Time Warping (DTW)?

Intuition: DTW finds the **optimal elastic alignment** between two time series by allowing non-linear warping of the time axis.

Given sequences $P = (p_1, \dots, p_m)$ and $Q = (q_1, \dots, q_n)$, construct a cost matrix $D \in \mathbb{R}^{m \times n}$:

$$D(i,j) = d(p_i, q_j) + \min \begin{cases} D(i-1,j) \\ D(i,j-1) \\ D(i-1,j-1) \end{cases}$$

$$\mathrm{DTW}(P,Q) = D(m,n)$$



DTW: Properties & Constraints

Algorithmic Properties

- Solved via dynamic programming in $O(mn)$
- Boundary conditions: path starts at $(1, 1)$, ends at (m, n)
- Monotonicity: indices never decrease along the path
- Continuity: no skipped indices

Strengths

- Handles phase shifts naturally
- Works on series of unequal length
- Well-studied in speech, ECG, finance

Limitations for Epidemic Curves

- Minimizes *total* cost — can dilate/compress the curve aggressively
- Hard to disentangle timing vs. magnitude
- Not a true metric (no triangle inequality)

What Is Fréchet Distance?

Intuition: Visualize walking your dog on a leash and while you walk along curve P, the dog walks along Q. Fréchet distance is the **minimum leash length** needed over all valid traversals.

Formally, for curves $P, Q : [0,1] \to \mathbb{R}^d$:

$$d_F(P, Q) = \inf_{\alpha,\beta} \max_{t \in [0,1]} d\big(P(\alpha(t)), Q(\beta(t))\big)$$

where $\alpha, \beta : [0,1] \to [0,1]$ are continuous, non-decreasing reparametrizations with $\alpha(0) = \beta(0) = 0$, $\alpha(1) = \beta(1) = 1$.

Discrete Version (our implementation):

$$d_F(P, Q) = \min_{C} \max_{(i,j) \in C} d(p_i, q_j)$$

where C is a valid monotone coupling path through the $m \times n$ grid.

Critical Distinction from DTW

- Fréchet minimizes the **maximum** paired distance
- DTW minimizes the **total sum**
- Fréchet is sensitive to the *worst-case* local mismatch

DTW vs. Fréchet: Side-by-Side

Property	**DTW**	**Fréchet Distance**
Optimization target	Minimize *sum* of distances	Minimize *maximum* distance
Sensitivity	Global shape similarity	Worst-case local mismatch
Algorithm	DP, $O(mn)$	DP, $O(mn)$
Interpretability	Accumulated mismatch	Bottleneck mismatch
Phase shift handling	Excellent	Good
Magnitude sensitivity	Moderate	High

Why Both?

Each method captures a different facet of curve dissimilarity!

Applying DTW & Fréchet to HSA–State Pairs

Setup

1. For each HSA h and its parent state s:
   - Extract weekly incidence vectors $\mathbf{p}_h$ (HSA) and $\mathbf{q}_s$ (state)
   - Align on the same 148-week grid
2. Compute $\texttt{DTW}(\mathbf{p}_h, \mathbf{q}_s)$ and $d_F(\mathbf{p}_h, \mathbf{q}_s)$ within each seasonal peak window
3. Extract derived metrics (timing, magnitude, composite)

Derived Metrics from DTW & Fréchet

Timing Metrics

- `signed_lag`: $\bar{i} - \bar{j}$ where i indexes HSA and j indexes state along the warping path
  - Positive $\Rightarrow$ HSA peaks *later*
  - Negative $\Rightarrow$ HSA peaks *earlier*
- `centroid_time_diff`: `hsa_centroid` $-$ `state_centroid` (mass-based timing)

Magnitude Metrics

- `weekly_mag_ratio`: `inc_hsa`/`inc_state` (week-by-week)
  - ¿1 $\Rightarrow$ HSA is consistently *larger*
  - ¡1 $\Rightarrow$ HSA is consistently *lower*
- `truth_mag_ratio`: ground-truth peak magnitude ratio

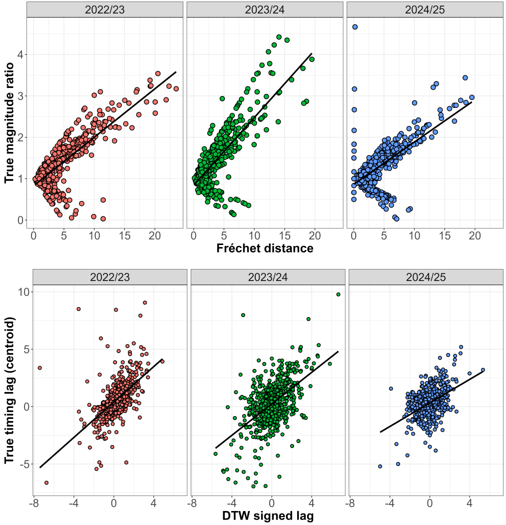
Validating Against Ground Truth

Truth Metrics (from raw ED data)

- `truth_time_diff`: actual week-of-peak difference between HSA and state
- `truth_mag_ratio`: ratio of actual peak incidence rates

Validation Strategy

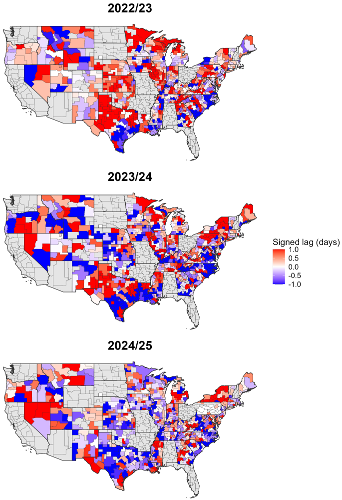
1. Compute correlation between metric vs. truth by peak window and season



Strong Correlations

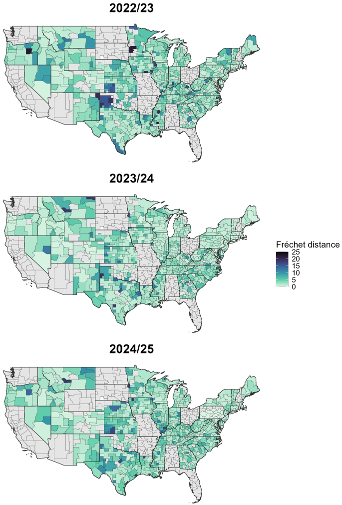
- `signed_lag` ∼ `truth_time_diff`
- `frechet` ∼ `truth_mag_ratio`

US Maps (Timing)



Interpretation

- Warm $\Rightarrow$ HSA peaks *after* state (lagging)
- Cool $\Rightarrow$ HSA peaks *before* state (leading)
- Geographic clusters emerge: Mountain West HSAs often lead; Midwest often lags
- Consistent spatial pattern across seasons suggests structural, not stochastic, divergence



Interpretation

- High `frechet` (darker) = worst-case local mismatch is large
- Certain areas show higher consistent Fréchet areas, like the top of Texas and Oklahoma region
- Spatial clustering hints at regional flu spread dynamics

Truth Metrics

Truth Metrics Are Simpler & Interpretable

- `truth_time_diff` is just the plain difference in week-of-peak
- `truth_mag_ratio` is the ratio of peak incidence rates is directly meaningful to epidemiologists
- These are the *ground truth* we validate against

So Why Bother?

1. Truth metrics only capture the **peak** while DTW/Fréchet capture the **full curve**
2. Multi-modal seasons: some HSAs have double peaks where "truth" timing is ambiguous
3. DTW/Fréchet generalize across curve shapes naturally

Limitations

DTW Limitations

- Can over-warp (pathological alignments at endpoints)
- Hard to understand intuitively at times
- Only considers one mapping possibility (deterministic)

Fréchet Limitations

- Sensitive to outlier spikes
- Assumes monotone traversal
- May confound timing and magnitude in a single number

Takeaway

Neither metric is strictly superior to truth metrics. They are *complementary* and most useful when ground truth peaks are unavailable or ambiguous, and most interpretable when validated against truth.

Probabilistic Framework

Current State

- Point estimates of DTW/Fréchet per HSA-season-window
- No uncertainty quantification on metric values

Why Probabilistic?

- ED data has inherent noise
- Seasonal epidemic curves are realizations of a stochastic process
- Decision-makers need *confidence intervals*, not just point estimates

Directions Under Consideration

1. **SoftDTW** from the `dtwclust` package along with extra code for additional metrics beyond cost
2. **Bootstraping distributions** by resampling weeks within a season to get CIs on `frechet` since there is no established method for a probabilistic version
3. **Composite metric formalization** by establishing weights for timing and magnitude components after extensive research

Composite Metric

Motivation

- Timing and magnitude are distinct epidemiological concepts
- Neither DTW nor Fréchet isolates them cleanly
- A composite could provide a single, interpretable divergence score

Candidate Form

$$\mathcal{C}(h,s) = w_1 \cdot \underbrace{|\texttt{signed\_lag}|}_{\text{timing}} + w_2 \cdot \underbrace{|\log \texttt{mag\_ratio}|}_{\text{magnitude}}$$

where w_1, w_2 could be learned from validation correlations.

Longer-Term Goals

- Extend to COVID-19 and RSV curves
- Identify structural predictors of divergence (urbanicity, geography, population size)

Summary

Our Work

1. Built an R pipeline for HSA–state epidemic curve comparison across 3 flu seasons
2. Implemented and compared DTW and discrete Fréchet distance
3. Derived timing and magnitude metrics
4. Validated against ground truth peak timing and magnitude ratios
5. Produced metric scatter plots and US choropleth maps

Key Findings

- Both metrics correlate with truth
- Geographic patterns in divergence are consistent across seasons
- Rural/small-population HSAs diverge most from state curves

Conclusion

DTW and Fréchet offer interpretable, validated alternatives to RMSE for epidemic curve comparison.